

Mid-term exam

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Content

The purpose of the mid term exam is to review the main topics we have reviewed so far:

- Basic image statistic
- Linear filters
- Feature extraction
- Image segmentation
- Classification using traditional ML methods

Image statistics

- Understand how we calculate an image histogram
- Understand how we can binarise an image
- Limitations of global thresholding

Linear filtering

Revise the definitions of linear convolutions in one and two dimensions

Remember the role of some specific filters: Gaussian, Sobel, Median, etc

Understand how a convolution affects the frequency information in an image

Feature extraction

- Revise how we extracted edges, blobs, and vessels
- Make you understand the different steps of the SIFT descriptor

Image segmentation

Core materials are the following algorithms:

- Image segmentation through k-means clustering
- The mean shift segmentation algorithm
- Superpixels
- Graph based clustering & normalised cuts

ML

- Revise the base concepts
- Dimensionality reduction methods
- Classification methods: logistic regression, SVMs, neural networks

What Students Will Learn from Lectures

You may present the content in different formats:

- Structure of the Course (session-by-session overview)
- Outline of the Course (main topics and progression)
- Theme/Objective/Goal (big-picture focus and learning outcomes)

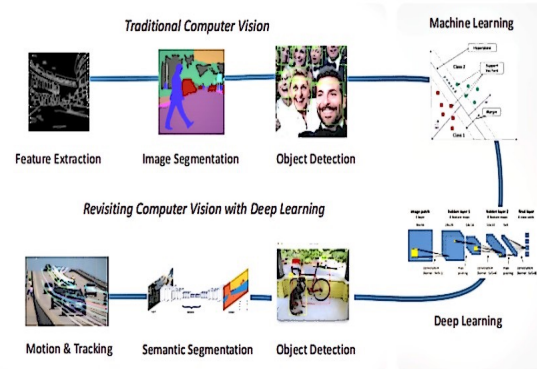
More on Critical Thinking: *the unique attribute of this course*



The overarching emphasis of this course: *critical thinking (CT)*:

- *Your competitive edge in professional world;*
- *Making you a more versatile and adaptive member of the society;*
- *Provide you clear reasoning of all important decisions in life;*
- *Making you a more logical person rather than a person living by the thoughts of others;*
- *The multi-faceted nature of CT: we use the topic of integrated circuit technology as a vehicle for practicing CT.*

Structure of the course



Recommended format for reference

Beyond the Lecture: Pathways to Research

You may present the value of research and the research course opportunities available, or refer to points such as:

Deepen knowledge and hands-on experience

Individual engagement with professor

Undertake supervised research project

Possibility of recommendation letters

2. Why Research

Research's significance and value

❖ The importance of research in academia and society: **Progress is driven by Research.**

Research Experience Benefits

❖ Skills to develop through research experiences (problem-solving skills, critical thinking, independent and collaborative working abilities, etc)—**transferable skills!**

Research Opportunity

❖ Individual engagement with professor, personalized guidance and support, undertake a supervised research project etc

Research Opportunities we offer

- Individual engagement with Professor,
- Personalized guidance and support,
- Undertake a supervised research project
- Support from highly skilled and trained teaching and research assistants
- Hands-on experience with numerical analysis, machine learning, coding and visualization
- Gain experience in MATLAB, python, numerical analysis



If you would like to include students' group photos from on-site programs, please check the attached email materials

Showcase of Student Research Excellence

You may present examples of student projects on this slide

Alternatively, you can connect each lecture's key concepts to student/potential projects as shown below (if preferred, please inform the coordinator, past students' manuscripts can be provided).

Introduction

Revenue

One-Sided

Two-Sided

Example of Student Project

1 Introduction

2 Revenue

3 One-Sided

4 Two-Sided

5 Example of Student Project

Introduction Revenue One-Sided Two-Sided Example of Student Project

TikTok II

- The paper considers user attitudes to different advertising channels:
 - product placement;
 - live-streaming;
 - push-ads.
- Students did a survey of 1119 respondents.
- Collected data on demographics, platform usage and reaction to various advertising channels (self-reported completed purchases).
- This design allows to make claims on which channel is more efficient for which group of users or product category.

Platform Pricing Search Direction Algorithms Pricing Student Projects

Potential Projects

- 1 Analysis of various fee structures in Armstrong (2006) framework – a project we have discussed before.
- 2 A project looked at whether a CEO gender affects adoption of AI by the company:
 - Several thousand observations: dataset of Chinese companies.
 - AI adoption is proxied by how often it is mentioned in the final report of the company.
 - Working hypothesis: having female CEO is associated with lower level of AI adoption, when controlled for company size, industry, etc.
- 3 Setting up a Q-learning algorithm and see how the price competition evolves under various market conditions. This project would require access to substantial computing power and good programming skills.